

COURSE: CHE 312: Transport Phenomena II

Fall Semester 2025

Department of Chemical Engineering

University of Illinois Chicago

Exam #3

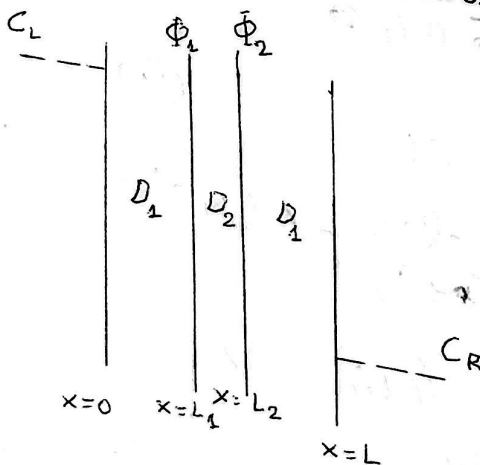
Duration: 70 minutes --- November 25, 2025

All Problems must be done

Problem 1: (30 points)

Consider steady state diffusion of the species A through three media arranged as shown in the attached diagram. Assume that diffusion is one-dimensional. At $x = 0$, $C_{A1} = C_L$, and at $x = L$, $C_{A3} = C_R$. In medium 2, there is a zeroth order chemical reaction through which the species A is consumed. Develop an expression for the concentration profiles across the three media.

(Bonus 10 points) Develop an expression for the steady state flux across the three media.



B.C at $x=0$ $C_{A1} = C_L$
 at $x=L$ $C_{A3} = C_R$

Medium 2 zeroth order reaction

$$\frac{d^2 C_A}{dx^2} = \frac{R}{D_A}$$

$$R = k_A$$

Medium 1 & Medium 3 $\rightarrow \frac{d^2 C_A}{dx^2} = 0$, $\frac{d C_A}{dx} = F$

Medium 2 $\frac{d^2 C_A}{dx^2} = \frac{k}{D_A}$

$\frac{d C_A}{dx} = \frac{k}{2D_A} x + A_0$

$C_A(x) = \frac{k}{2D_A} x^2 + A_0 x + B_0$

at $x=0$

$$C_A(0) = C_L$$

Medium 1 $\frac{dC_A}{dx} = F_1$, $C_A = F_1 x + J_1$

$$J_1 = C_L$$

$$\therefore C_A(x) = F_1 x + C_L \quad \text{for } 0 \leq x \leq L_1$$

Similarly Medium 2 $\rightarrow C_A(x) = F_2 x + C_R$

Medium 2

at $x = L_1$

$$C_A(L_1) = C_A(L_1^+) \Rightarrow F_1 L_1 + C_L = \frac{K}{2D_A} (L_1)^2 + A L_1 + B \quad \text{--- (1)}$$

at $x = L_2$

$$C_A(L_2) = C_A(L_2^+) \Rightarrow \frac{K}{2D_A} (L_2)^2 + A L_2 + B = F_1 L_2 + C_R \quad \text{--- (2)}$$

$$\text{(1)} \Rightarrow B = F_1 L_1 + C_L - \frac{K}{2D_A} (L_1)^2 - A L_1 \quad \text{--- (3)}$$

$$\text{(2)} \Rightarrow B = F_1 L_2 + C_R - \frac{K}{2D_A} (L_2)^2 - A L_2 \quad \text{--- (4)}$$

$$\text{(3)} - \text{(4)} \Rightarrow F_1 (L_1 - L_2) + C_L - C_R - \frac{K}{2D_A} ((L_1)^2 - (L_2)^2) - A (L_1 - L_2) = 0$$

$$\therefore A = \frac{-F_1 (L_1 - L_2) - C_L + C_R + \frac{K}{2D_A} ((L_1)^2 - (L_2)^2)}{(L_1 - L_2)}$$

$$B = F_1 L_1 + C_L - \frac{K}{2D_A} (L_1)^2 - \left(\frac{-F_1 (L_1 - L_2) - C_L + C_R + \frac{K}{2D_A} ((L_1)^2 - (L_2)^2)}{(L_1 - L_2)} \right) (L_1 - L_2)$$

$$C_A(x) = \frac{k}{2DA} x^2 + Ax + B$$

$$C_A(x) = \frac{k}{2DA} x^2 + \left[\frac{-F_1(L_1 - L_2) - C_L + C_R + \frac{k}{2DA} (L_1^2 - L_2^2)}{L_1 - L_2} \right] x$$

$$+ F_1 L_1 + C_L - \frac{k}{2DA} (L_1)^2 + \left[\frac{-F_1(L_1 - L_2) - C_L + C_R + \frac{k}{2DA} (L_1^2 - L_2^2)}{L_1 - L_2} \right] L_2$$

Problem 2

Case 2

$$J_{\text{int}} = K_1$$

$$J_{\text{ext}} = K_{-1} C_B$$

$$\frac{d^2 C_A}{dy^2} = \frac{J_{\text{ext}} - J_{\text{int}}}{D} \quad \frac{d^2 C_B}{dy^2} = \frac{J_{\text{int}} - J_{\text{ext}}}{D}$$

Boundary Conditions

$$C_A(0) = C_A^0 \quad \& \quad C_B(0) = C_B^0$$

$$\frac{dC_A}{dy}(L) = 0 \quad \& \quad \frac{dC_B}{dy}(L) = 0$$

$$\frac{d^2 C_A}{dy^2} = \frac{K - K_{-1} C_B}{D} \Rightarrow \frac{dC_A}{dy} = \frac{K - K_{-1} C_B}{D} y + A_1$$

$$A_1 = \frac{-K - K_{-1} C_B L}{D}$$

$$\therefore \frac{dC_A}{dy} = \frac{K - K_{-1} C_B (y - L)}{D}$$

$$C_A(y) = \frac{K - K_{-1} C_B}{2D} (y - L)^2 + A_2$$

Using B.C

$$C_A(0) = C_A^0 \Rightarrow A_2 = C_A^0 - \frac{K - K_{-1} C_B}{2D} L^2$$

$$C_A(y) = C_A^0 - \frac{K - K_{-1} C_B}{2D} (L^2 - y^2)$$

Problem 2: (70 points)

Consider the reversible conversion of species A into B through the stoichiometric reactions $A \xrightleftharpoons[k_{-1}]{k_1} B$ in a stationary liquid film of thickness L. Determine C_A and C_B throughout the film, if

(35 Points) the surface at $y = L$ is assumed to be impermeable and unreactive, and the concentrations of A and B at $y = 0$ are specified as C_{A0} and C_{B0} . The forward and reverse homogeneous reactions follow zeroth order kinetics

(35 Points) the surface at $y = L$ is assumed to be impermeable and unreactive, and the concentrations of A and B at $y = 0$ are specified as C_{A0} and C_{B0} . The forward and reverse homogeneous reactions follow zeroth and first order kinetics, respectively

Case 1

$$\text{Rate} = k_f$$

$$\therefore \frac{dC_A}{dy} = -k_f$$

$$\frac{dC_B}{dy} = k_f$$

$$\text{at } y = 0, C_A(0) = C_{A0}$$

$$\text{at } y = 0, C_B(0) = C_{B0}$$

$$R_{\text{forward}} = -k_1 C_A$$

$$R_{\text{reverse}} = k_{-1} C_B$$

$$r_A = -k_1 C_A + k_{-1} C_B, \quad r_B = k_1 C_A - k_{-1} C_B$$

$$\frac{d}{dy} \left(D_A \frac{dC_A}{dy} \right) = r_A$$

$$\frac{d}{dy} \left(D_B \frac{dC_B}{dy} \right) = r_B$$

Assuming $D_A = D_B = D$

We get $\frac{d}{dy} \left(D \frac{dC_A}{dy} \right) = -k_1 C_A + k_{-1} C_B \Rightarrow \text{for A}$

For B, $\frac{d}{dy} \left(D \frac{dC_B}{dy} \right) = k_1 C_A - k_{-1} C_B$

at $y = 0, L$ $\frac{dC_A}{dy} = 0$ & $\frac{dC_B}{dy} = 0$

we can write, $D \frac{d^2 C_A}{dy^2} = -\frac{k_T}{D} C_A + \frac{\Delta k}{D} C_B$ $\Delta k = k_1 - k_{-1}$

$D \frac{d^2 C_B}{dy^2} = \frac{k_T}{D} C_B - \frac{\Delta k}{D} C_A$ $k_T = k_1 + k_{-1}$
